ÖRNEK = $\sqrt[3]{2\sqrt[3]{4}} = \sqrt[3]{\sqrt[3]{2^{2\cdot3}\cdot4}} = \sqrt[3\cdot2\cdot3]{2^6\cdot2^2} = \sqrt[18]{2^8} = \sqrt[9]{2^4} = 2^{\frac{4}{9}}$

ÖRNEK = $a>0$ olmak üzere $(\sqrt[5]{a^2})^4$ ifadesinin eşiti nedir?

$\Rightarrow (\sqrt[5]{a^2})^4 = \sqrt[5]{a^8} = \sqrt[5]{a^5\cdot a^3} = a\sqrt[5]{a^3}$

ÖRNEK = $\sqrt{(-4)^2} - \sqrt[3]{(-2)^3} + \sqrt[4]{(-2)^4} + \sqrt[5]{3^5} = |-4| - (-2) + |-2| + 3 = 4+2+2+3 = 11$ //

ÖRNEK = $a<0$ olmak üzere $\sqrt{a^2} + \sqrt{(-a)^2} - \sqrt{4a^2}$ ifadesinin eşiti nedir?

$= a<0 \Rightarrow |a| = -a \qquad \Rightarrow a>0,\ |-a| = -a$

$= \sqrt{a^2} + \sqrt{(-a)^2} - \sqrt{4a^2} = |a| + \sqrt{(-a)^2} - \sqrt{(2a)^2}$

$= |a| + |-a| - |2a|$

$= -a + (-a) - (-2a)$

$= -a - a + 2a \quad = 0$ //

ÖRNEK = $a>0$ ve $b<0$ ise $\sqrt{(b-a)^2} - \sqrt{(2a-b)^2}$ ifadesinin eşiti nedir?

$\Rightarrow b-a<0$ ve $2a-b>0 \Rightarrow \sqrt{(b-a)^2} - \sqrt{(2a-b)^2} = |b-a| - |2a-b|$

$= -(b-a) - (2a-b)$

$= -b + a - 2a + b$

$= -a$ //

ÖRNEK = $\frac{2}{\sqrt{6}} = \frac{2\sqrt{6}}{6} = \frac{\sqrt{6}}{3}$

$(\sqrt{6})$

$= \frac{2\sqrt{3}}{\sqrt{5}-\sqrt{3}} = \frac{2\sqrt{3}(\sqrt{5}+\sqrt{3})}{\sqrt{5}^2-\sqrt{3}^2} = \frac{2\sqrt{15}+6}{5-3} = \frac{2(\sqrt{15}+3)}{2} = \sqrt{15}+3$

$(\sqrt{5}+\sqrt{3})$

ÖRNEK = $A=\sqrt{5}-\sqrt{3}$ ve $B=\sqrt{5}+\sqrt{3}$ olduğuna göre $\frac{A}{B}+\frac{B}{A}+2$ ifadesi neye eşittir.

$\frac{A}{B}+\frac{B}{A}+2 = \frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}+\sqrt{3}} + \frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}} + 2$

$(\sqrt{5}-\sqrt{3})\quad(\sqrt{5}+\sqrt{3})$

$\frac{(\sqrt{5}-\sqrt{3})^2}{\sqrt{5}^2-\sqrt{3}^2} + \frac{(\sqrt{5}+\sqrt{3})^2}{\sqrt{5}^2-\sqrt{3}^2} + 2$

$\frac{5-2\sqrt{15}+3+5+2\sqrt{15}+3}{2} + 2$

$8+2 = 10$

ÖRNEK = $x < 0 < y$ ise $\frac{|2x-y| + |2y-x|}{x+y}$ ifadesi neye eşittir?

$\Rightarrow x < 0 < y \;=\; 2x < 0 \quad 2y > 0 \;=\; 2x - y < 0 \quad 2y - x > 0$

$\Rightarrow \frac{|2x-y| - |2y-x|}{x+y} = \frac{-(2x-y)-(2y-x)}{x+y} = \frac{-2x+y-2y+x}{x+y} = \frac{-(x+y)}{x+y} = -1$ //

ÖRNEK = $\sqrt{25} \in R$, $\sqrt{0} \in R$, $\sqrt[4]{81} \in R$, $\sqrt[5]{7} \in R$, $\sqrt[3]{0} \in R$, $\sqrt{-3} \notin R$, $\sqrt[3]{-3} \in R$

ÖRNEK = $\sqrt{0,64} + \sqrt{0,04} - \sqrt[3]{0,027}$ $\Rightarrow \sqrt{\frac{64}{100}} + \sqrt{\frac{4}{100}} - 3\sqrt{\frac{27}{1000}}$

$\Rightarrow \sqrt{\left(\frac{8}{10}\right)^2} + \sqrt{\left(\frac{2}{10}\right)^2} - 3\sqrt{\left(\frac{3}{10}\right)^3}$

$\Rightarrow \frac{8}{10} + \frac{2}{10} - \frac{3}{10}$

$\Rightarrow \frac{7}{10} = 0,7$ //

ÖRNEK = $\frac{\sqrt{75} - \sqrt{27} + \sqrt{12}}{\sqrt{108} + \sqrt{48}} = \frac{\sqrt{5^2 \cdot 3} - \sqrt{3^2 \cdot 3} + \sqrt{2^2 \cdot 3}}{\sqrt{6^2 \cdot 3} + \sqrt{4^2 \cdot 3}} = \frac{5\sqrt{3} - 3\sqrt{3} + 2\sqrt{3}}{6\sqrt{3} + 4\sqrt{3}} = \frac{4\sqrt{3}}{10\sqrt{3}} = \frac{2}{5}$ //

ÖRNEK = $\sqrt{2} \cdot \left(\sqrt{18} + \sqrt{50} - \sqrt{32} + \sqrt{8}\right) = \sqrt{36} + \sqrt{100} - \sqrt{64} + \sqrt{16} = 6 + 10 - 8 + 4 = 12$ //

ÖRNEK = $\frac{\sqrt[5]{16}}{\sqrt[5]{\frac{1}{2}}} = \sqrt[5]{\frac{16}{\frac{1}{2}}} = \sqrt[5]{32} = \sqrt[5]{2^5} = 2$

ÖRNEK = $\sqrt{25 - 25x^2} - \sqrt{64 - 64x^2} = \sqrt{25(1-x^2)} - \sqrt{64(1-x^2)} = 5\sqrt{1-x^2} - 8\sqrt{1-x^2}$

$\Rightarrow -3\sqrt{1-x^2}$

ÖRNEK = $\sqrt[5]{\sqrt{\sqrt{\sqrt[3]{\sqrt{2}}}}} = \sqrt[5 \cdot 2 \cdot 2 \cdot 3 \cdot 2]{2} = \sqrt[120]{2}$

ÖRNEK = $\dfrac{-9^{-2}\cdot 25^{-1}}{125^{-1}\cdot 27^{-1}} = ?$ $\Rightarrow \dfrac{-(3^2)^{-2}\cdot(5^2)^{-1}}{(5^3)^{-1}\cdot(3^3)^{-1}} = \dfrac{-3^{-4}\cdot 5^{-2}}{5^{-3}\cdot 3^{-3}} = -\dfrac{5^{3-2}}{3^{4-3}} = -\dfrac{5}{3}$

ÖRNEK = $\dfrac{x^{a+1}}{x^{a-2}}\cdot\left(\dfrac{x^{-a+2}\cdot x^{2a+3}}{x}\right)$ ifadesinin eşiti nedir?

$\Rightarrow x^{a+1-a+2}\cdot\left(x^{-a+2+2a+3-1}\right)^{-1} = x^3\cdot\left(x^{a+4}\right)^{-1} = x^3\cdot x^{-a-4} = x^{3-a-4} = x^{-a-1}$

ÖRNEK = $\left(\dfrac{a^x}{a^y}\right)^{x-y}\cdot\left(\dfrac{a^y}{a^x}\right)^{x-y}$ ifadesinin eşiti nedir?

1. yol $\Rightarrow \left(\dfrac{a^x}{a^y}\right)^{x-y}\cdot\left(\dfrac{a^y}{a^x}\right)^{x-y} = \left(a^{x-y}\right)^{x-y}\cdot\left(a^{y-x}\right)^{x-y}$

$= \left(a^{x-y}\cdot a^{y-x}\right)^{x-y} = \left(a^{x-y+y-x}\right)^{x-y} = \left(a^0\right)^{x-y} = 1^{x-y} = 1$ //

2. yol $\Rightarrow \left(\dfrac{a^x}{a^y}\right)^{x-y}\cdot\left(\dfrac{a^y}{a^x}\right)^{x-y} = \left(\dfrac{a^x}{a^y}\cdot\dfrac{a^y}{a^x}\right)^{x-y} \Rightarrow \left(\dfrac{a^x}{a^y}\cdot\dfrac{a^y}{a^x}\right)^{x-y} = 1^{x-y} = 1$

ÖRNEK = $3^x = 2$ olduğuna göre $27^{x+1} - 81^x$ ifadesinin değeri kaçtır?

$\Rightarrow \left(3^3\right)^{x+1} - \left(3^4\right)^x = 3^{3x+3} - 3^{4x}$

$= \left(3^x\right)^3\cdot 3^3 - \left(3^x\right)^4$

$= 2^3\cdot 3^3 - 2^4 = 200$

ÖRNEK = $x-2=y$ olduğuna göre $|x-y|+|y-x|$ ifadesi neye eşittir?

$\Rightarrow x-2=y \Rightarrow x-y=2$ $\Rightarrow |x-y|+|y-x| = |2|+|-2| = 2+2 = 4$ //

$\Rightarrow y-x=-2$

ÖRNEK = $x\in R$ ve $-2<x<4$ olduğuna göre $|x+2|+|x+4|$ ifadesinin eşiti nedir?

$\Rightarrow -2<x<4 \Rightarrow x+2>0$ ve $x+4<0$

$|x+2|+|x-4| = x+2-(x-4 = \cancel{x}+2-\cancel{x}+4 = 6$ //

ÖRNEK = $4^3 = 4 \cdot 4 \cdot 4 = 64$

$(-2)^4 = (-2)\cdot(-2)\cdot(-2)\cdot(-2) = 16$

$(-5)^3 = (-5)\cdot(-5)\cdot(-5) = -125$

ÖRNEK = $\Rightarrow 25^0 = 1$ $\quad \Rightarrow \left(-\frac{2}{5}\right)^{-1} = -\frac{5}{2}$

$\Rightarrow (-9)^0 = 1$ $\quad \Rightarrow \left(\frac{3}{4}\right)^{-2} = \left(\frac{4}{3}\right)^2 = \frac{16}{9}$

$\Rightarrow -2014^0 = -1$ $\quad \Rightarrow (-2)^{-5} = \left(-\frac{1}{2}\right)^5 = -\frac{1}{32}$

ÖRNEK = $-\left[\left(-\frac{1}{2}\right)^{-1}\right]^4 = -\left(-\frac{1}{2}\right)^{-4} = -(-2)^4 = -2^4 = -16$

ÖRNEK = $-4^5 + 3\cdot 4^5 - 7\cdot 3^4 - 3\cdot 3^4 - 4^5 = 4^5\cdot(-1+3-1) + 3^4\cdot(-7-3)$

$= 4^5 - 10\cdot 3^4$

ÖRNEK = $\frac{5^{80} + 5^{85} + 5^{90}}{5^{85} + 5^{80} + 5^{75}} = x$ ise $\frac{x}{125}$ ifadesinin eşiti kaçtır?

Çözüm = $x = \frac{5^{80}\cdot(1+5^5+5^{10})}{5^{75}\cdot(5^{10}+5^5+1)} = \frac{5^{80}}{5^{75}} = 5^{80-75} = 5^5 //$ $\quad = x \quad \frac{5^5}{125} = \frac{5^5}{5^3} = 5^{5-3} = 5^2 = 25 //$

ÖRNEK = $\frac{14^n + 6^n + 2^n}{7^n + 3^n + 1}$ ifadesi neye eşittir?

$\Rightarrow \frac{(7\cdot 2)^n + (3\cdot 2)^n + 2^n}{7^n + 3^n + 1} = \frac{7^n\cdot 2^n + 3^n\cdot 2^n + 2^n}{7^n + 3^n + 1} = \frac{2^n\cdot(7^n + 3^n + 1)}{7^n + 3^n + 1} = 2^n$

ÖRNEK = $(0{,}027)^{\frac{5}{3}}\cdot 10^5$ işleminin sonucu neye eşittir?

$\Rightarrow (27\cdot 10^{-3})^{\frac{5}{3}}\cdot 10^5 = (3^3)^{\frac{5}{3}}\cdot(10^{-3})^{\frac{5}{3}}\cdot 10^5$

$\Rightarrow 3^5\cdot 10^{-5}\cdot 10^5 = 3^5\cdot 10^{-5+5} = 3^5\cdot 10^0 = 3^5 //$